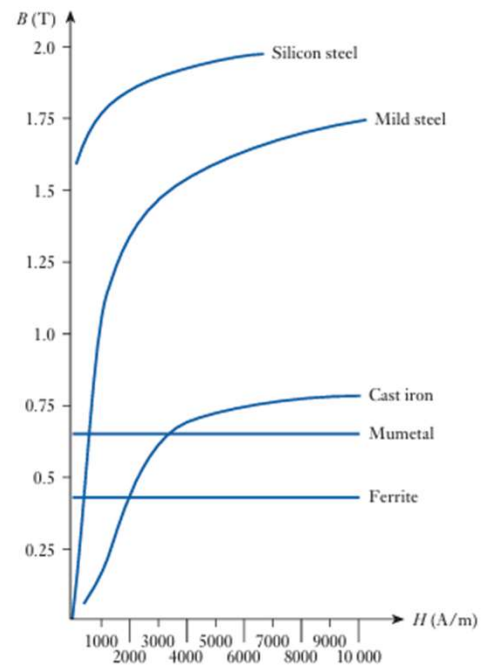


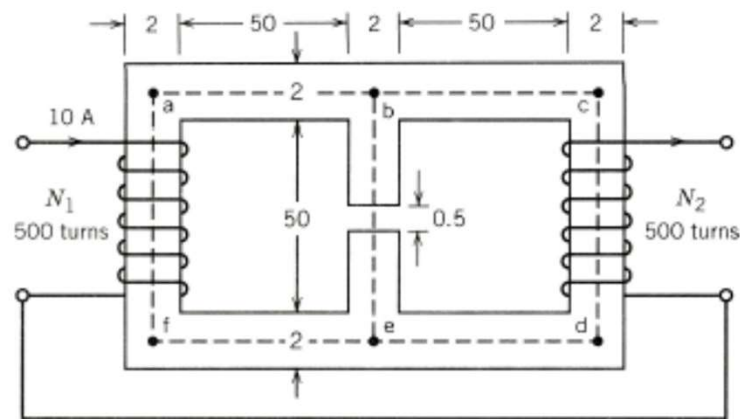
Magnetisation characteristics

$$B = \mu_r \mu_0 H$$

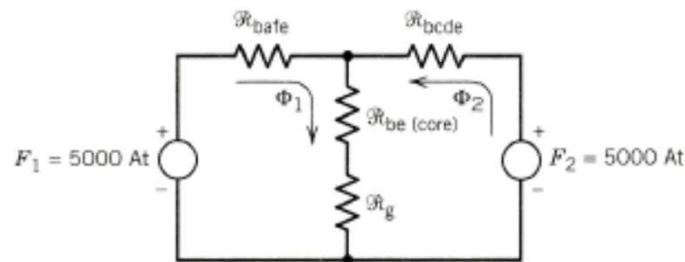


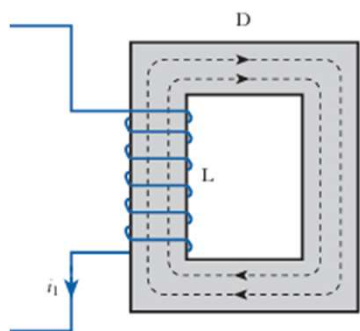
Electric circuit		Magnetic circuit	
Ohm's law: $V = IR$		'Ohm's law': $F = \Phi S$	
Electromotive force (e.m.f.) V	volt	Magnetomotive force (m.m.f.) $F (=NI)$	ampere (turn)
Current I	amp	Magnetic flux Φ	weber
Conductivity σ	siemens	Permeability μ	henry per metre
Resistance $R = l/\sigma A$	ohm	Reluctance $R = l/\mu A$	ampere (turn) per weber
Electric field strength E	volt per metre	Magnetic field strength H	ampere (turn) per metre
Current density $\mathcal{J}$	amps per m ²	Magnetic flux density B	tesla

In the magnetic circuit of Fig. El.3a, the relative permeability of the ferromagnetic material is 1200. Neglect magnetic leakage and fringing. All dimensions are in centimeters, and the magnetic material has a square cross-sectional area. Determine the air gap flux, the air gap flux density, and the magnetic field intensity in the air gap.



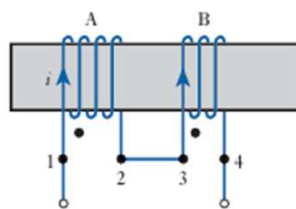
(a)





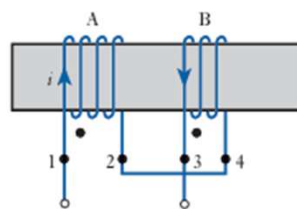
$$e = L \cdot \frac{di}{dt}$$

$$L = 4\pi \times 10^{-7} \times \frac{AN^2}{l} \text{ henrys}$$



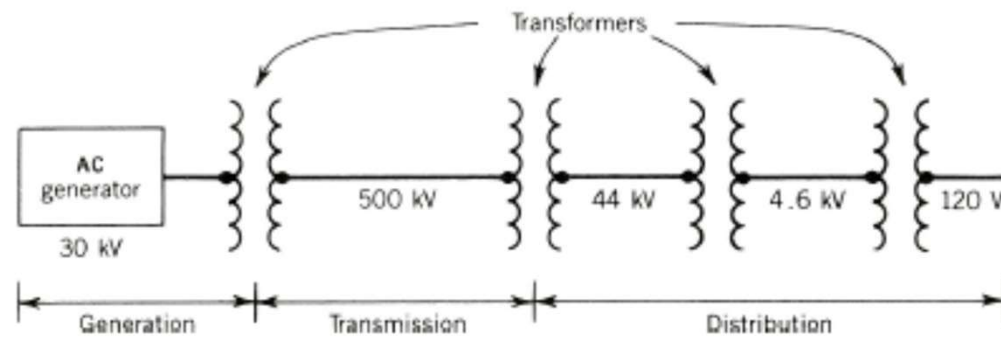
$$(L_A + L_B + 2M) \cdot \frac{di}{dt}$$

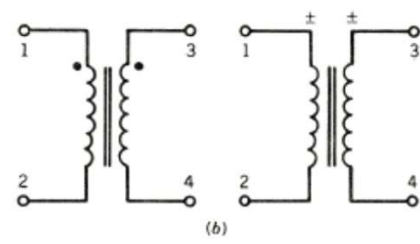
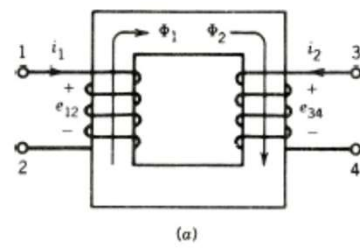
$$L_1 = L_A + L_B + 2M$$

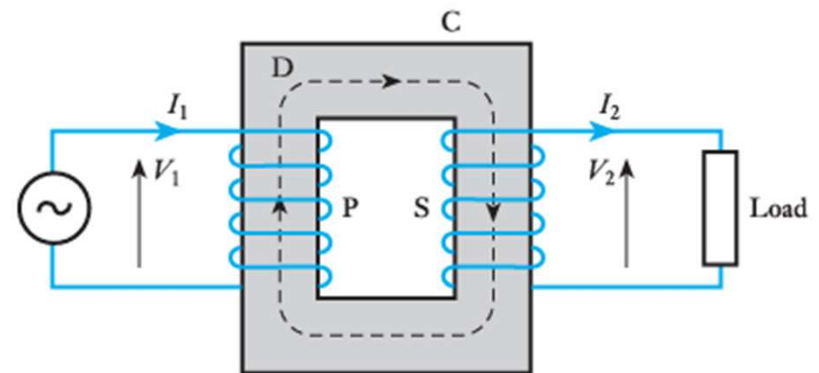
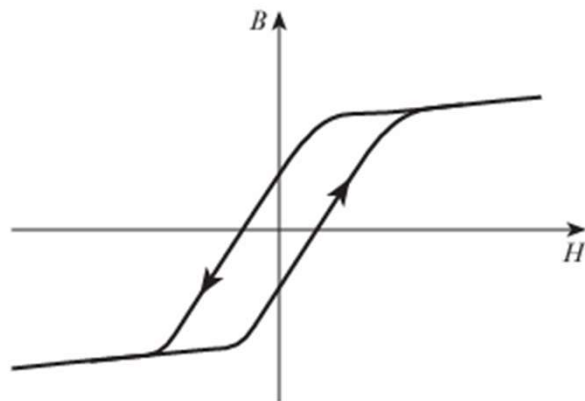


$$L_A \cdot \frac{di}{dt} + L_B \cdot \frac{di}{dt} - 2M \cdot \frac{di}{dt}$$

$$L_A + L_B \pm 2M$$







Core losses:
Hysteresis and
Eddy current losses

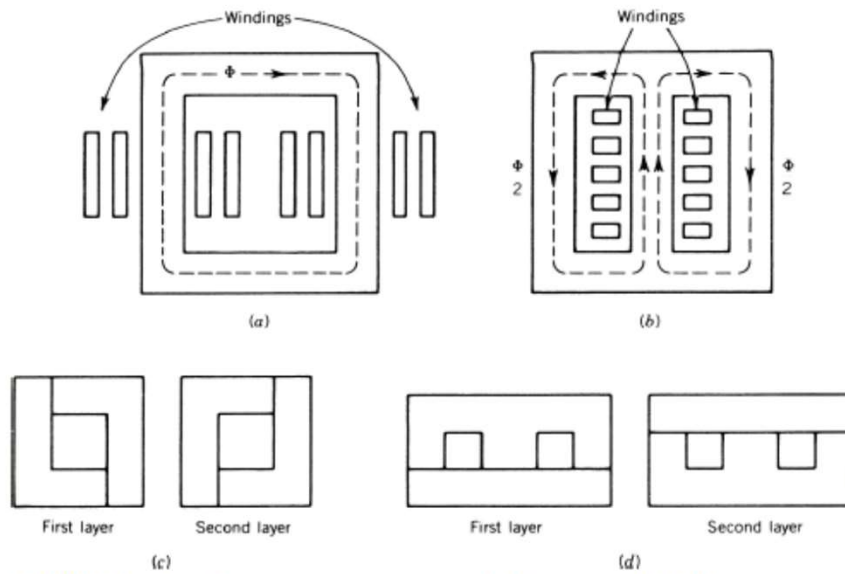
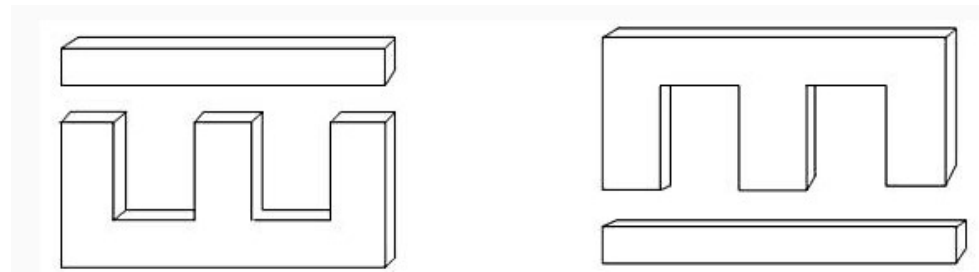
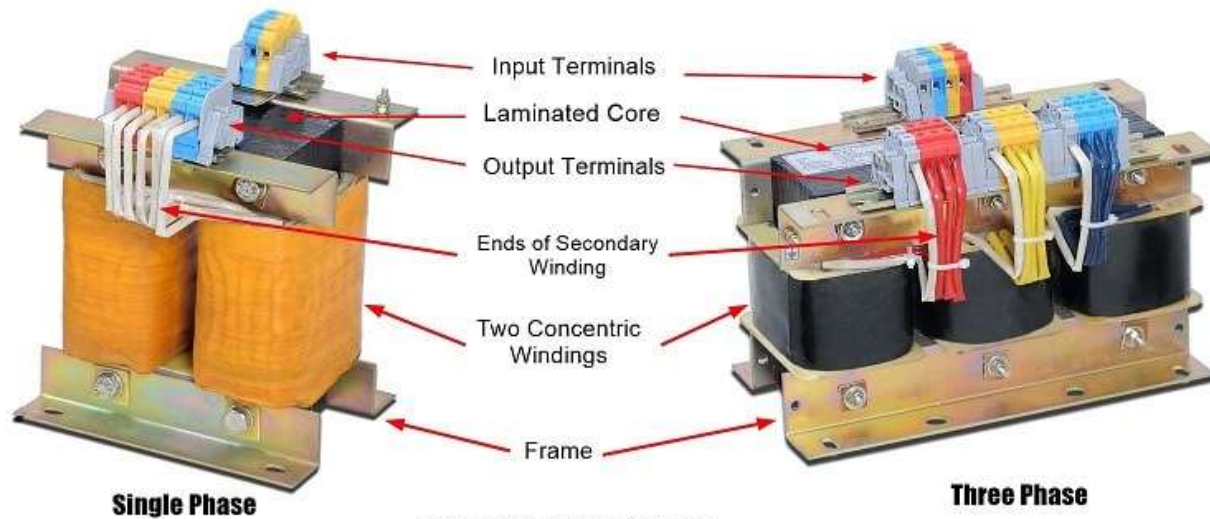


FIGURE 2.1 Transformer core construction. (a) Core-type, (b) Shell-type, (c) L-shaped lamination, (d) E-shaped lamination.

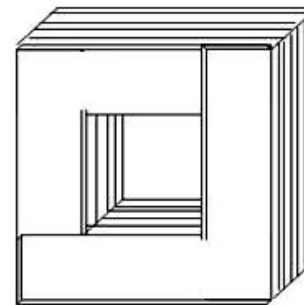
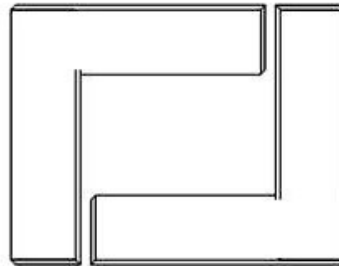


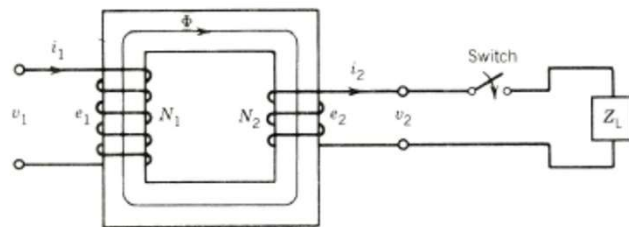
Silicon steel
laminations





Core Type Transformer





$$v_1 = e_1 = N_1 \frac{d\Phi}{dt}$$

$$v_2 = e_2 = N_2 \frac{d\Phi}{dt}$$

$$\frac{v_1}{v_2} = \frac{N_1}{N_2} = a$$

$$N_1 i_1 - N_2 i_2 = \text{net mmf} = 0$$

$$N_1 i_1 = N_2 i_2$$

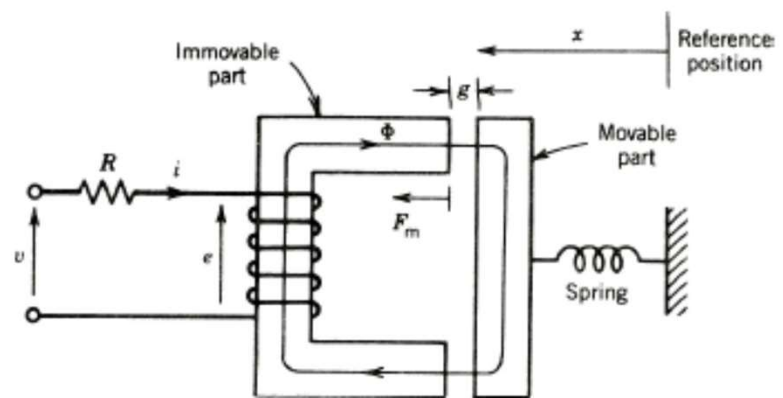
$$\frac{i_1}{i_2} = \frac{N_2}{N_1} = \frac{1}{a}$$

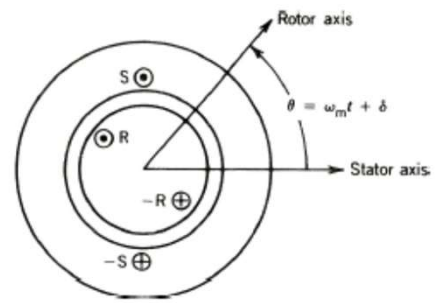
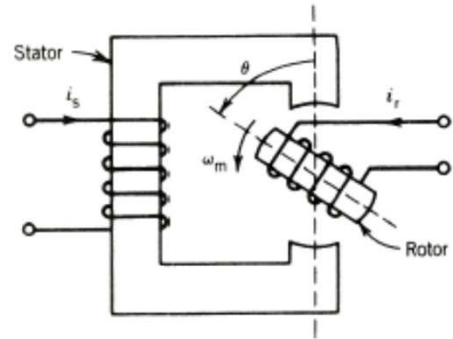
$$\frac{V_1}{V_2} = \frac{N_1}{N_2} = a$$

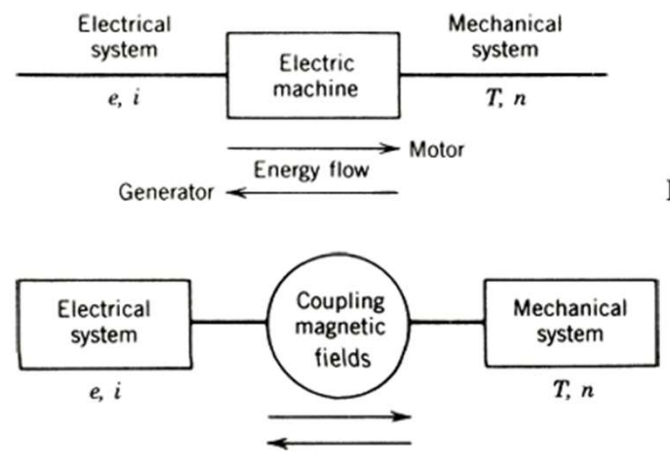
$$\frac{I_1}{I_2} = \frac{N_2}{N_1} = \frac{1}{a}$$

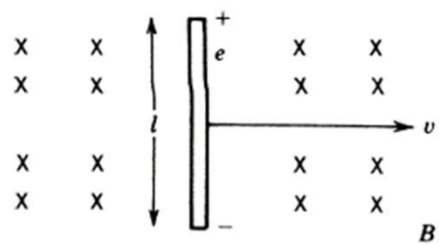
$$V_1 I_1 = V_2 I_2$$

↑ ↑
 input output
 volt-amperes volt-amperes



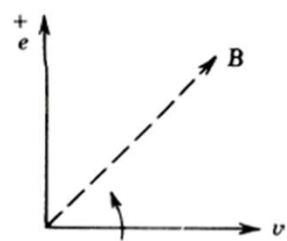






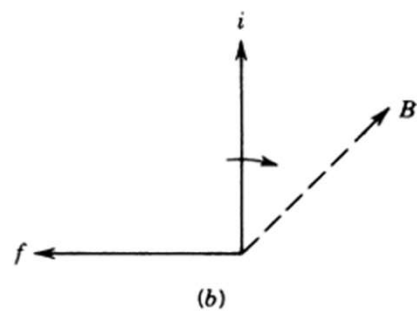
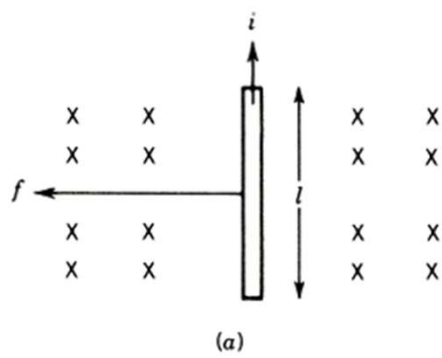
X indicates B into the paper

(a)

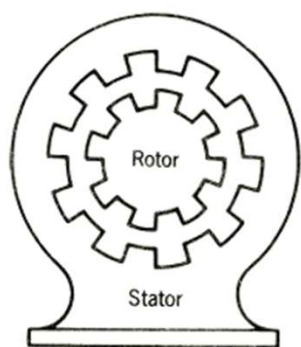


(b)

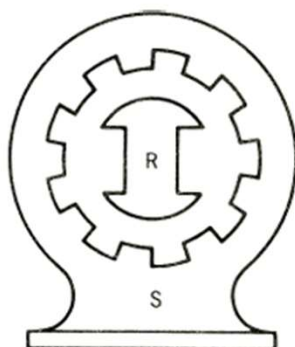
$$e = Blv$$



$$f = Bli$$

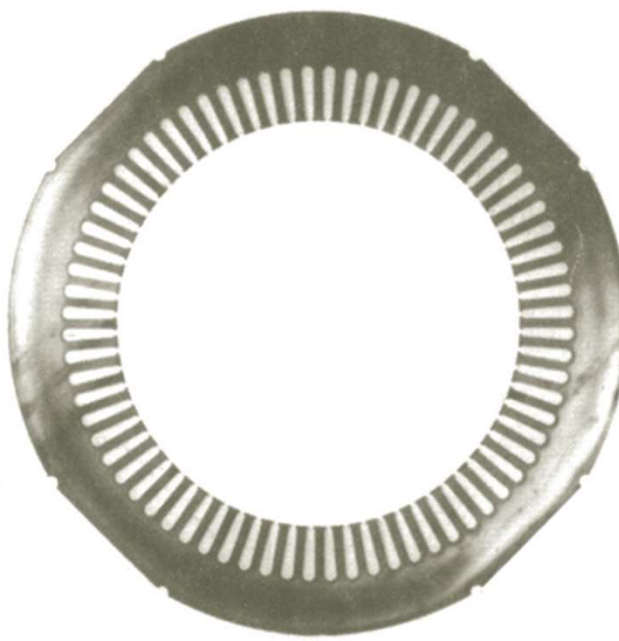


(a)

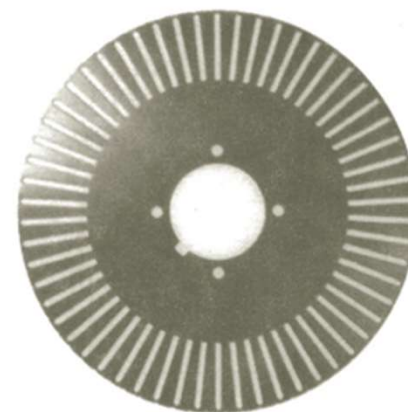


(b)

Courtesy of Westinghouse Canada Inc.



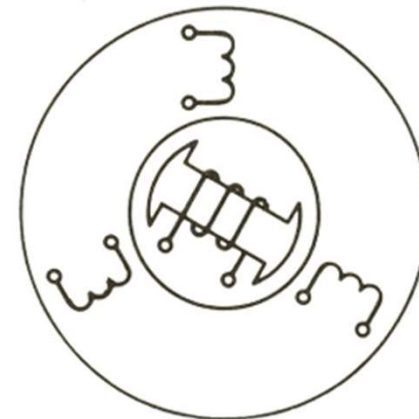
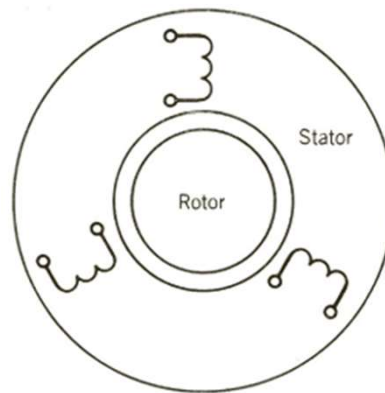
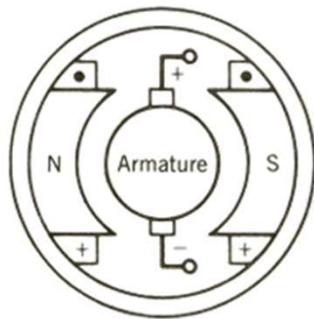
(a)



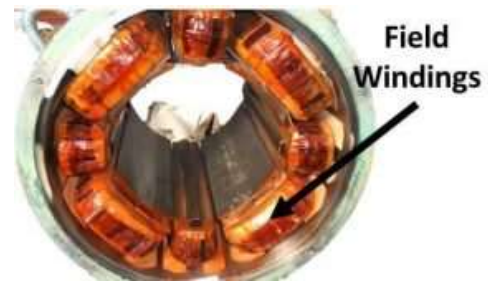
(b)

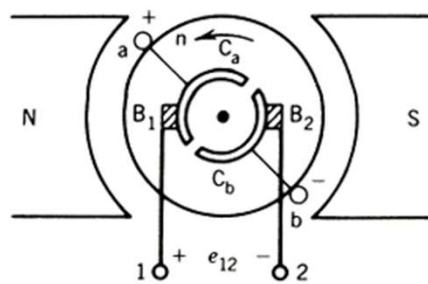
FIGURE 4.6 Laminations. (a) Stator. (b) Rotor.

	Field winding	Armature winding	
DC machine	Stator-DC excitation	Rotor-AC voltages induced from DC excitation	Rectification done through mechanical commutator
Induction motor	Stator (AC excitation)	Stator (AC)	Voltage induced in rotor
Synchronous generator	Rotor (DC excitation)	Stator (AC voltage induced)	Rotor connected through slip rings



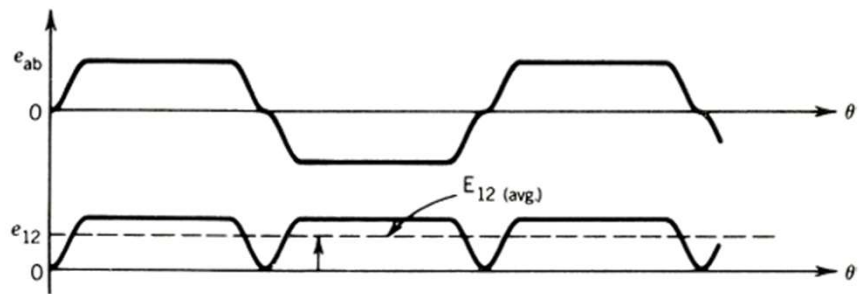
DC machine:



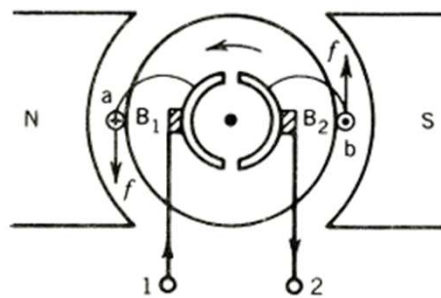


C_a, C_b -commutator segment

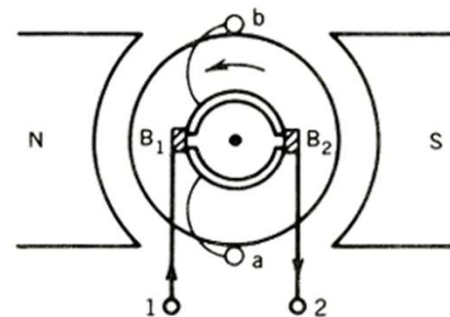
(a)



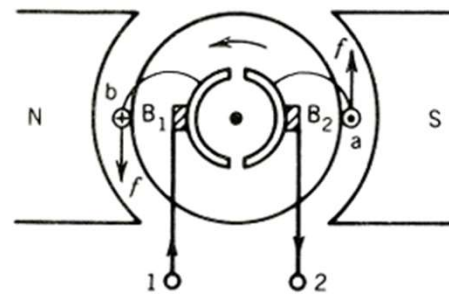
(b)

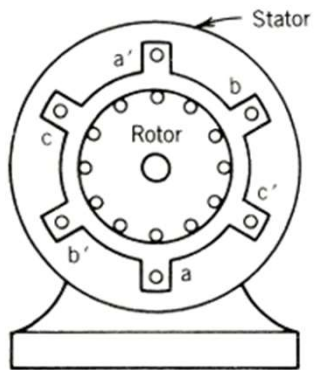


(a)

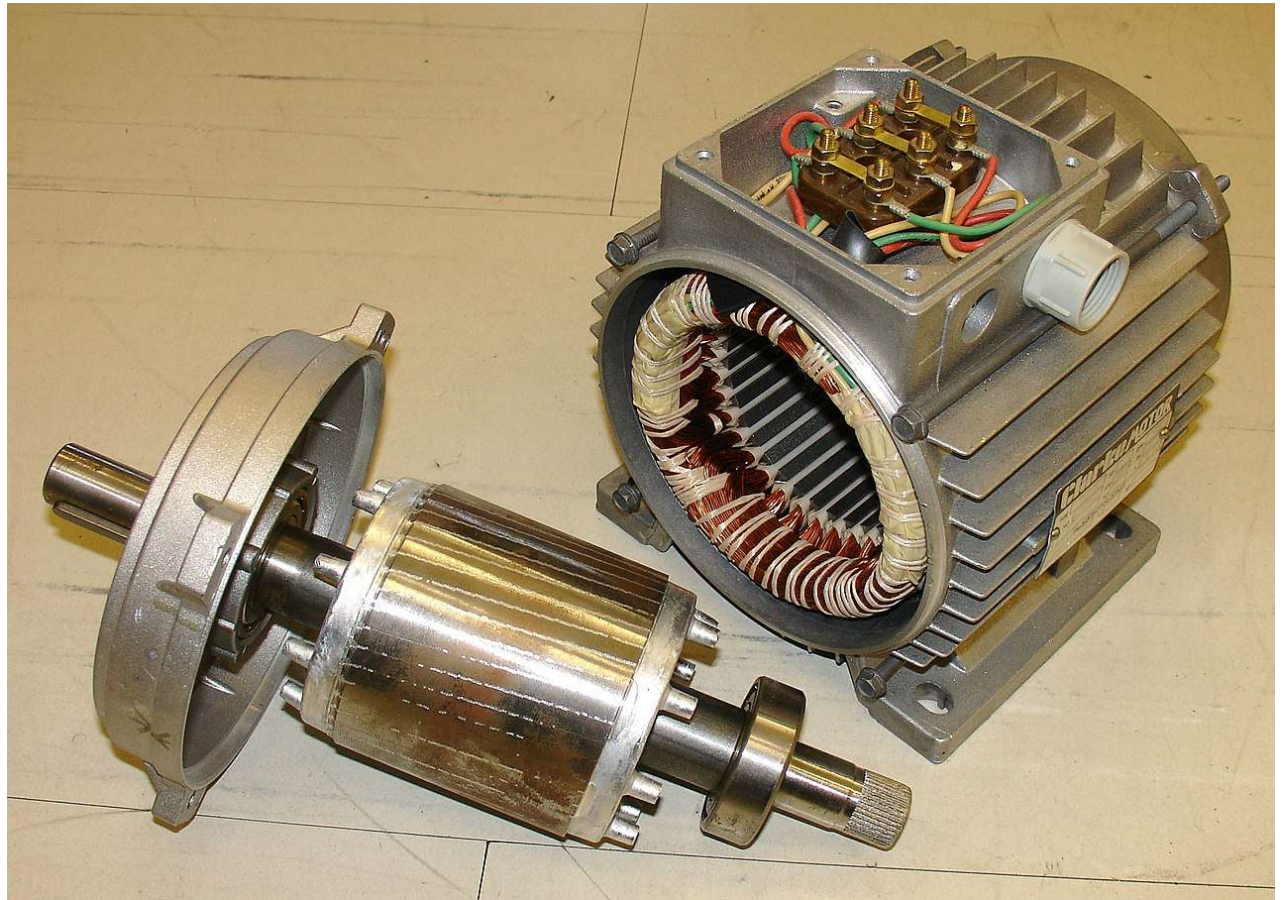


(b)

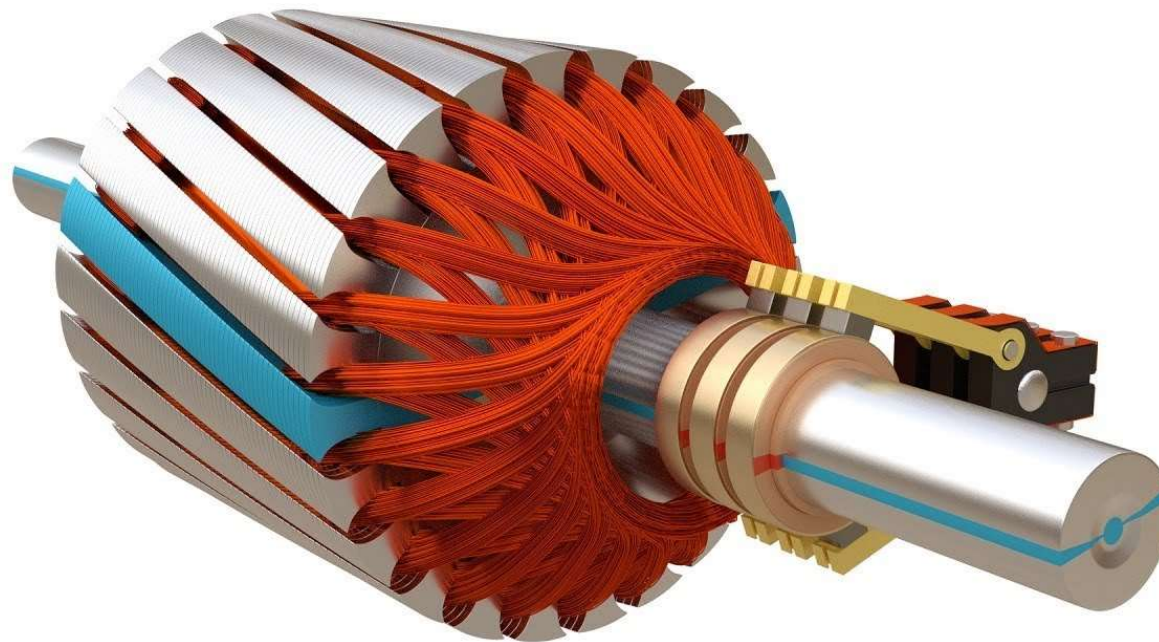


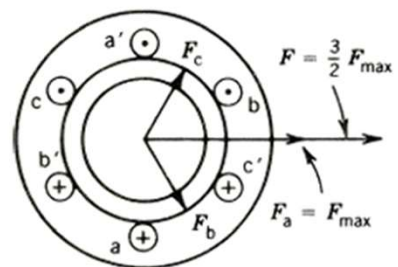
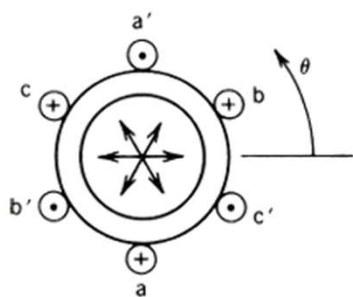


Squirrel cage
induction machine

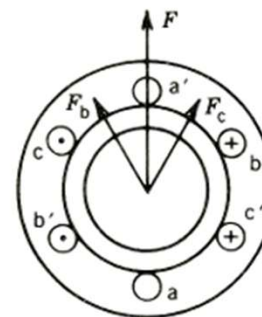


Slip ring induction machine

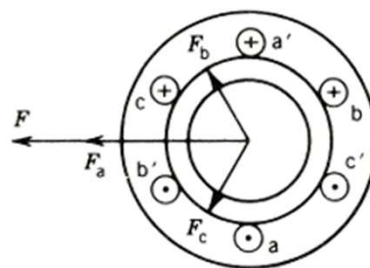
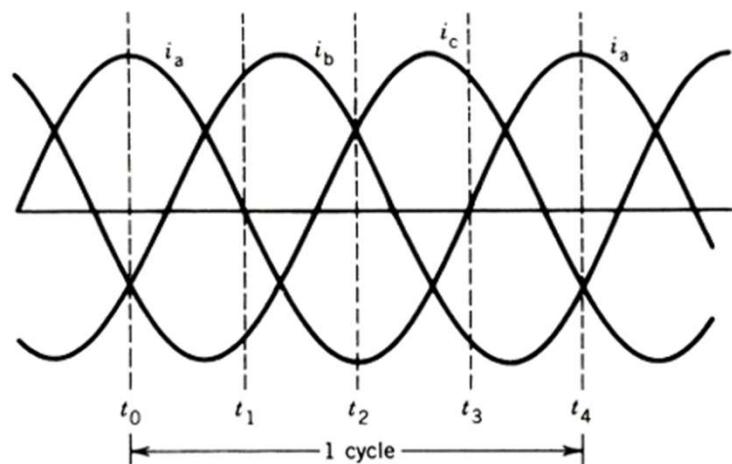




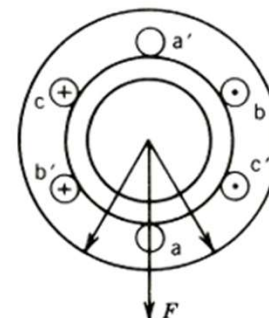
(a) $t = t_0 = t_4$



(c) $t = t_1$

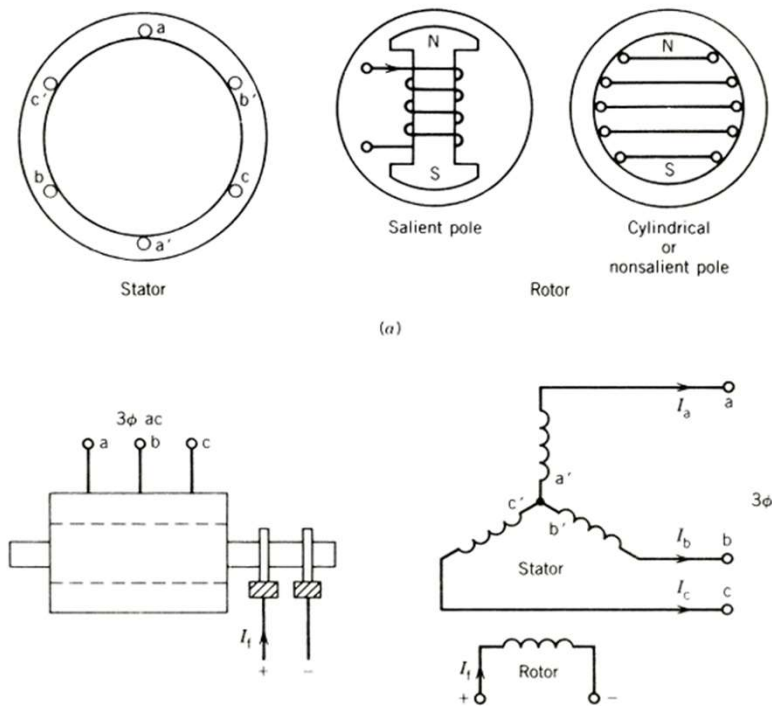


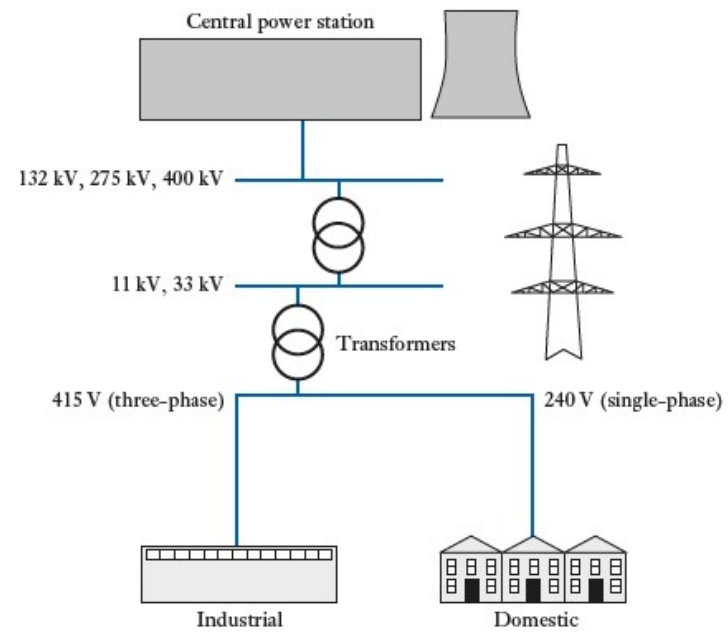
(d) $t = t_2$



(e) $t = t_3$

Synchronous generator





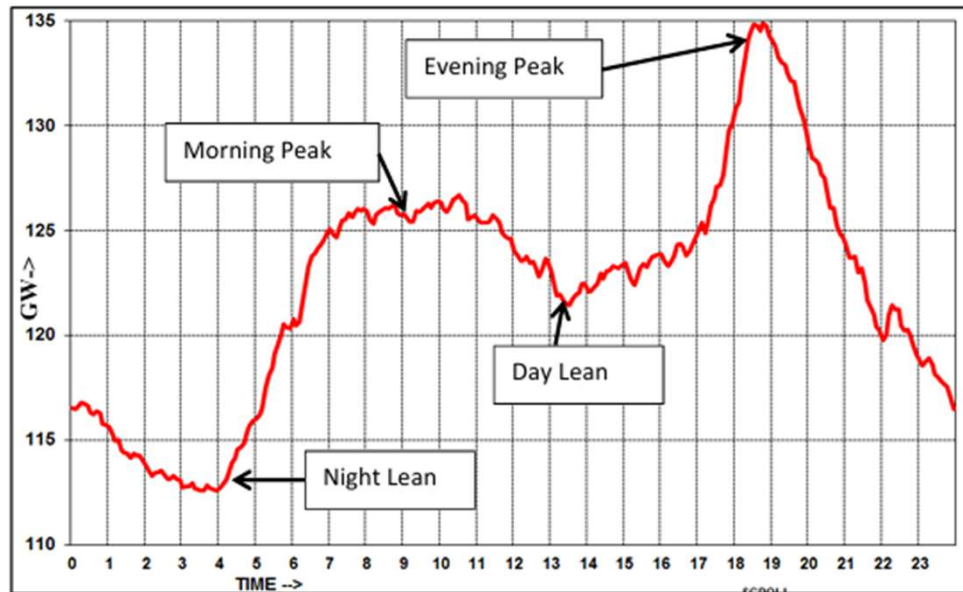


Figure 1: Typical All India Load Curve



Power System Operation Corporation Ltd.

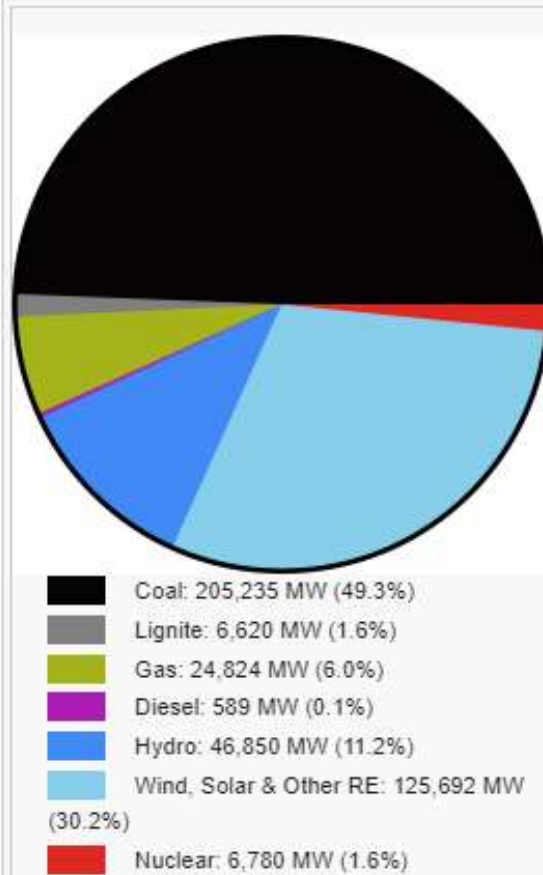
2016

Breakdown of Installed Generation Capacity as on 31/03/2023

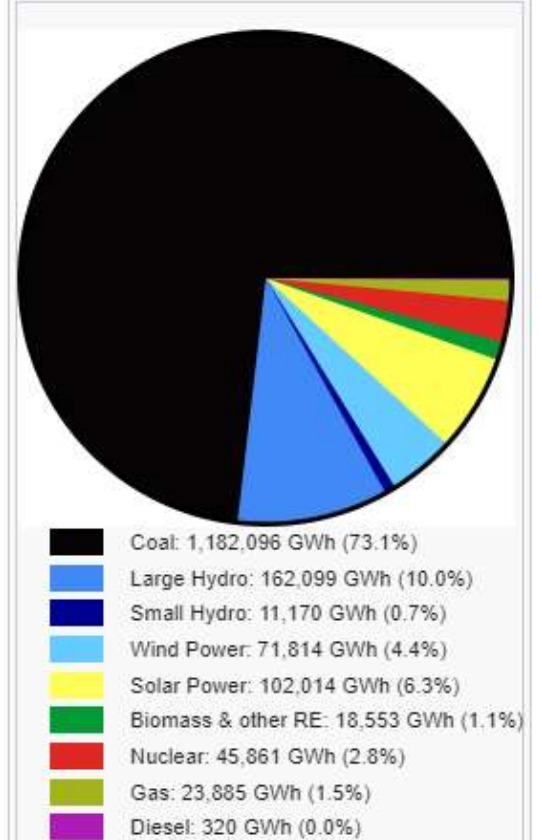
Source	Installed Capacity (MW)	% of Share in Total
Fossil Fuels (Total)	237,269	57%
Coal		49.3%
Lignite	6,620	1.6%
Gas	24,824	6.0%
Diesel	589	0.1%
Non-Fossil Fuels (Total)	178,790	43%
Hydro	46,850	11.3%
Wind	42,633	10.2%
Solar	66,780	16.1%
Bio Mass Power/Cogen	10,248	2.5%
Waste to Energy	554	0.1%
Small Hydro	4,944	1.2%
Nuclear	6,780	1.6%
Total Installed Capacity	416,059	100%

Wikipedia

Installed capacity by source in the utility sector as on 12 June 2023^[44]



Electricity generation (utility sector) by source in India in FY 2022-2023



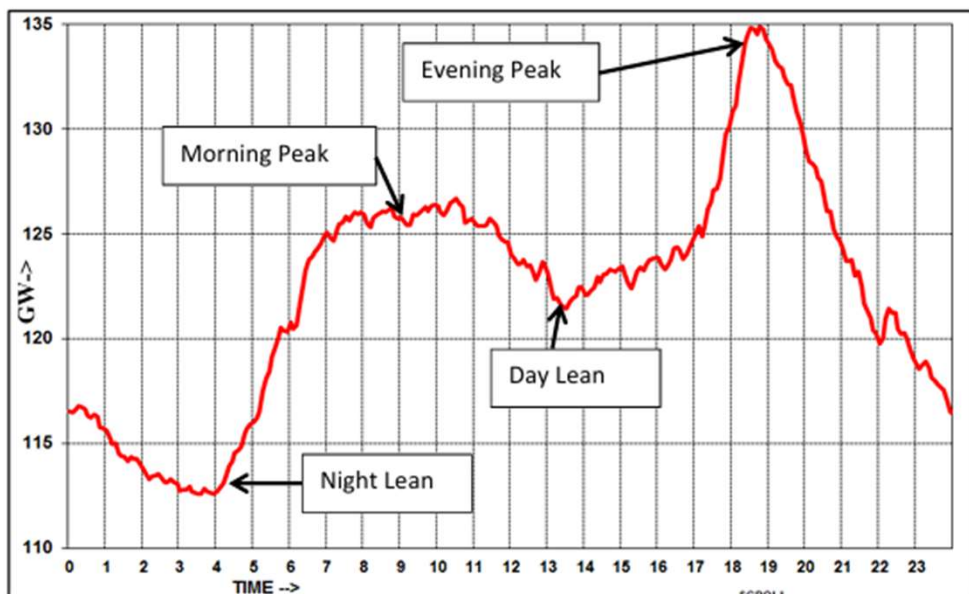
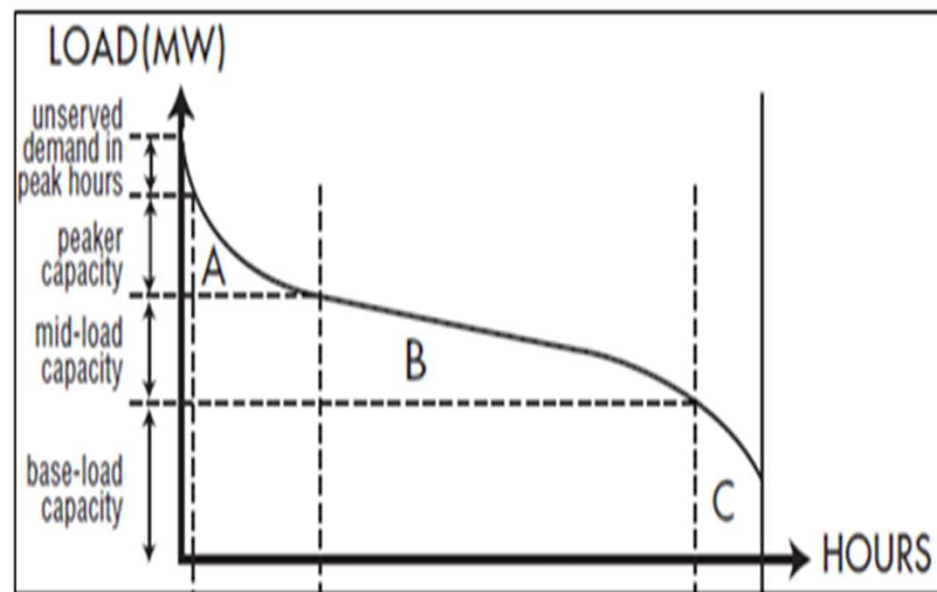


Figure 1: Typical All India Load Curve



Power System Operation Corporation Ltd.

2016

- Capacity factor=

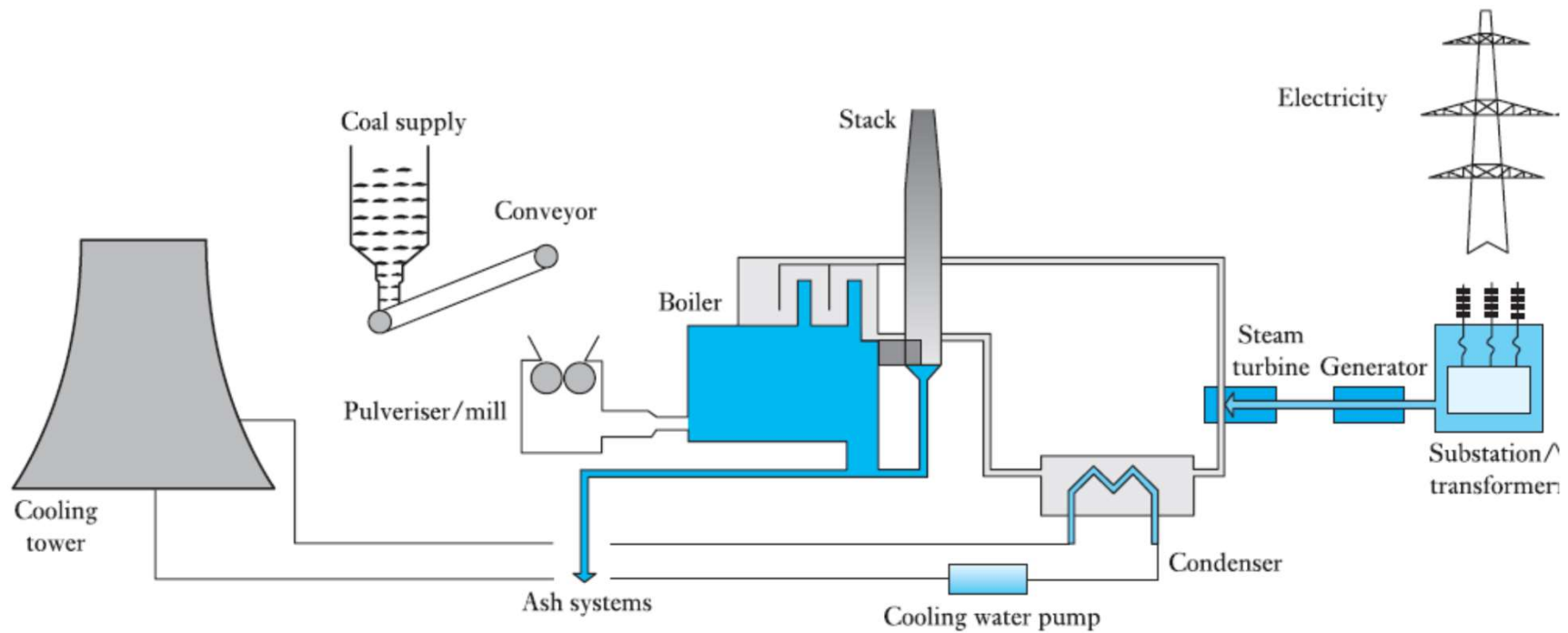
$$\frac{\text{Actual energy produced}}{\text{Energy it can produce when operating at its full capacity}}$$

Table 39.1 Energy conversion efficiencies

Low entropy High entropy ← → CONVERSION TO						
CONVERSION FROM	Mechanical	Gravitational potential	Electrical	Radiation (light)	Chemical	Thermal
Mechanical			99% (generator)			100% (brakes)
Gravitational potential	90% (water turbine)		85% (hydroelectric station)			
Electrical	95% (electric motor)	80% (pumped storage)		5% (filament bulb) 20% (fluorescent)	75% (battery)	100% (coil in kettle)
Radiation (light)			12% (PV solar cell)		0.6% (photosynthesis)	100% (solar furnace)
Chemical	30% (animals)		90% (dry cell)	15% (chemical laser)		88% (power station boiler)
Thermal	50% (steam turbine)		8% (thermocouple)			

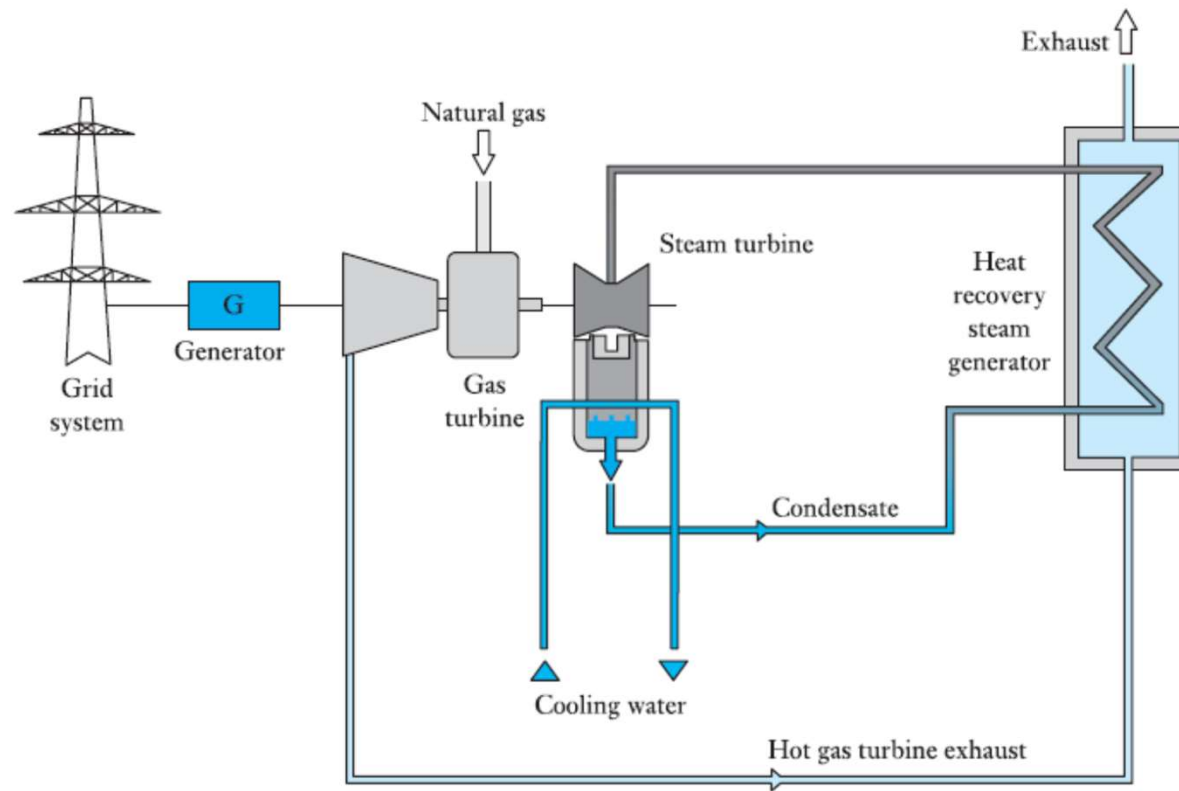
Thermal power plants

- Coal fired power plant

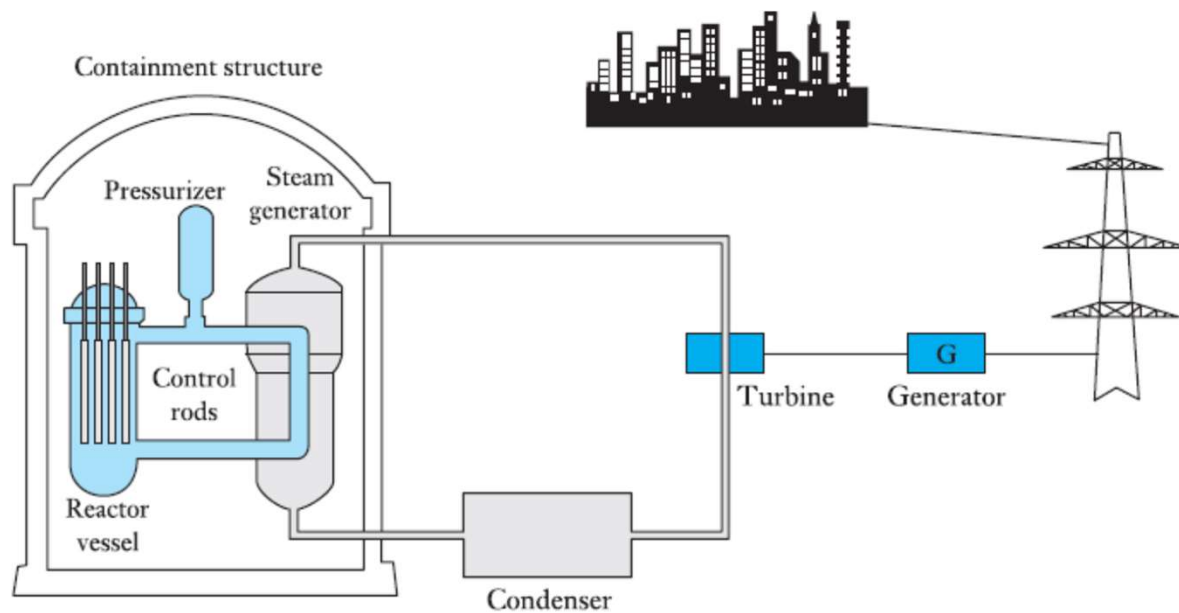


Thermal power plants

- Combined Cycle gas turbine plant



Thermal plant: PWR



Hydro Electric Power plant

